

Problem 9.5

Given that $i^{(m)} = 0.1844144$ and $d^{(m)} = 0.1802608$. Find m .

Solution

For nominal rates convertible m times per year, the relationship between the nominal interest rate and nominal discount rate is

$$i^{(m)} = d^{(m)} \left(1 + \frac{i^{(m)}}{m} \right).$$

Substituting the given values,

$$0.1844144 = 0.1802608 \left(1 + \frac{0.1844144}{m} \right).$$

Divide both sides by 0.1802608:

$$\frac{0.1844144}{0.1802608} = 1 + \frac{0.1844144}{m}.$$

Solving for m ,

$$\begin{aligned} \frac{0.1844144}{m} &= \frac{0.1844144}{0.1802608} - 1, \\ m &= \frac{0.1844144}{\frac{0.1844144}{0.1802608} - 1}, \\ m &\approx 8.003343431. \end{aligned}$$

Since m must be an integer number of conversion periods per year,

$$\boxed{m = 8}.$$